

CHAPTER 12

CHOICE AND MOMENTUM

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What proportion of a person's behavior is likely to be devoted to a response of social or clinical significance, whether it be desirable or undesirable? And how persistent is that response likely to be in different circumstances? This chapter will review basic research on the allocation of behavior between alternatives under constant conditions, including cases where only one alternative is well defined. It will also review research on the persistence of responding when behavior is disrupted by a change of conditions. Finally, it will discuss some applications of these findings in clinical situations.

The allocation of responding between two or more well-defined alternatives—hereafter termed *choice*, without implying that organisms actively select and initiate their own behavior—and the rate of responding when only one alternative is defined, are both encompassed by a single formulation of behavior in relation to relative rates of reinforcement known as *the matching law* (Herr-

stein, 1970). The matching law describes behavior under constant, stable conditions, but the real world is anything but constant. The next question is how reinforcement schedules affect resistance to change—characterized, metaphorically, as behavioral momentum—when conditions are altered. Research suggests that steady-state choice and response rates depend primarily on response-reinforcer relations, whereas resistance to change depends primarily on stimulus-reinforcer relations (Nevin, 1992b). The clinical implications of this distinction are especially important.

CHOICE AND MATCHING

Paradigms

The classic study of free-operant choice behavior was conducted by Herrnstein (1961), who trained food-deprived pigeons to peck at either of two lighted keys for food reinforcers. Food

was scheduled intermittently according to two variable-interval (VI) schedules that operated concurrently and independently. One arranged reinforcers for the first peck on one key after an unpredictable time averaging t_1 seconds, and the other arranged reinforcers for pecks on the other key after times averaging t_2 seconds. Each VI value can be translated into a reinforcer rate: for example, a VI schedule with an average of 40 seconds between reinforcers arranges 90 reinforcers per hour. Throughout this chapter, VI schedule values will be specified as *reinforcer rates*.

In Herrnstein's (1961) experiment, a pair of VI schedules remained in effect until performance stabilized—usually about 30 daily sessions. Over the course of the experiment, different pairs of schedules, chosen so that the sum of the reinforcer rates programmed by the two schedules was constant, were arranged in successive conditions. When reinforcers could be obtained only after a brief delay from the first peck at either key following a peck at the other key—a so-called changeover delay or COD—the results were strikingly orderly: the proportion of pecks at each key approximately equalled the proportion of reinforcers obtained from that key. The data, shown in Figure 12-1, are well described by a simple equation, known as the matching law:

$$B_1/(B_1 + B_2) = r_1/(r_1 + r_2), \quad (1)$$

where B_1 and B_2 represent the number of pecks emitted on keys 1 and 2, and r_1 and r_2 represent the number of reinforcers obtained by pecks on keys 1 and 2. This result and subsequent extensions provided the basis for Herrnstein's (1970) formulation of matching to relative reinforcement as a general law of behavior.

There are two features of concurrent VI VI schedules that must be appreciated in relation to the matching result. First, the probability of reinforcement for switching to an alternative increases with time spent away from it. Thus, as a subject continues to respond to the richer of the two schedules, at some time the probability of reinforcement for switching to the leaner schedule will be higher than for staying with the richer

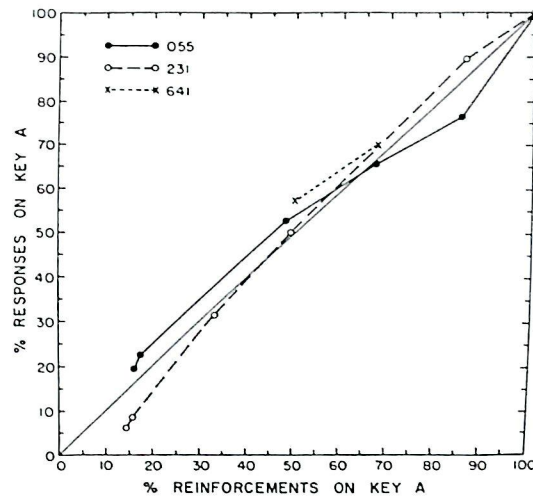


Figure 12-1. The proportion of responses made to one of two keys as a function of the proportion of food reinforcers received on that key, as arranged by concurrent VI VI schedules for three pigeons. The diagonal line indicates exact matching. From Herrnstein (1961).

schedule, thus encouraging responding to both alternatives. No such contingency operates in concurrent variable-ratio (VR) schedules: if the schedules differ, one alternative always has a higher probability of reinforcement than the other, and subjects usually respond exclusively to the richer alternative (Herrnstein & Loveland, 1975)—a result that is trivially compatible with Equation 1 because either B_1 and r_1 or B_2 and r_2 are zero.

Second, the high probability of reinforcement for switching on concurrent VI VI schedules makes switching very likely. When there was no COD in Herrnstein's (1961) experiment, the birds switched back and forth between the keys several thousand times in each session and distributed their pecks about equally between the two keys. The COD delayed reinforcement after each switch, thereby reducing switching rates and promoting independence of the two schedules. The role of the COD in concurrent-schedule performance is discussed at length in a comprehensive review of the matching law by Davison and McCarthy (1988).

An earlier experiment by Findley (1958), also using pigeons as subjects, arranged food reinforcers for pecks at a single key according to concurrent VI VI schedules. However, reinforcers could be obtained from only one schedule at a time, depending on the color of the food key. The pigeons could alternate between colors by pecking a second, "switching" key, and thus obtain all reinforcers from both schedules on the food key. The procedure is particularly well suited to measuring time spent engaged in one or the other alternative, defined by key-color time. Findley's paradigm has come to be known as the *switching-key* procedure, whereas Herrnstein's has been termed the *two-key* procedure for studying concurrent operants. By and large, the results of these two procedures are similar.

A third paradigm, introduced by Autor (1960/1969), separated choice between schedules of reinforcement from the behavior that obtains food reinforcers. In Autor's procedure, which is termed *concurrent-chain schedules*, identical concurrent VI VI schedules were arranged on two white keys, defining the initial links of the concurrent chains. The initial-link schedules were arranged as in Herrnstein's (1961) study, but without a COD. When a reinforcer was available on one key, a peck at that key produced a change of key color—to red, for example—instead of food, and the other key was darkened. Further pecks on the red key were reinforced with food according to a VI schedule. The red key and its correlated food schedule defined one terminal link; a different terminal link—correlated with green, for example—was produced by pecks at the other key during the initial links. The concurrent initial links were reinstated after food was presented; hence, the terminal links differed in the average delay to food. The reciprocal of delay is termed *immediacy*, and average immediacy is equivalent to rate of food presentation.

Concurrent chains may be viewed as two-key concurrent schedules of conditioned reinforcement, where the conditioned reinforcing values of the terminal-link stimuli depend on the immediacies of food reinforcement arranged by the VI schedules in their presence. Autor varied the ter-

minal-link schedules over successive conditions and found that the distribution of initial-link key pecks matched the relative immediacy (or rate) of food in the terminal links. Thus, early research suggested that the matching law provided an excellent account of food-reinforced choice behavior of pigeons in three related paradigms.

Generality of Matching

The matching result for choice has remarkable generality. For example, matching to the relative duration of food provided by equal concurrent VI VI schedules has been observed with pigeons in the two-key and switching-key paradigms (Brownstein, 1971; Catania, 1963). Matching has also been obtained with rats, monkeys, and humans in experiments employing a variety of different responses and reinforcers, including electrical brain stimulation, drugs, and points exchangeable for money (for review, see Davison & McCarthy, 1988; de Villiers, 1977; Williams, 1988).

An experiment by Conger and Killeen (1974) is of special interest because it employed human subjects in a social setting. College undergraduates were studied one at a time in conversation with three "plants" designated E_1 , E_2 , and E_3 . E_3 engaged the subject in conversation, and E_1 and E_2 provided verbal reinforcers such as "that's a good point" whenever a VI schedule, cued separately for E_1 and E_2 in such a way as to be hidden from the subject, had timed out and the subject had made some comment to E_1 or E_2 . In the first half of the 30-minute session, E_1 provided 70% of the scheduled reinforcers while E_2 provided 30%; their roles were reversed for the second half of the session. The principal measure was the proportion of time that the subject spent talking to E_1 or E_2 , who were, in effect, playing the roles of the two keys in Herrnstein's (1961) experiment described previously. When the results were pooled for all five subjects, Equation 1 was confirmed almost exactly. The data were more variable than those of Figure 12-1, probably because exposure to the schedules was so brief, but they strongly suggest that the matching law, which was identified for hungry pigeons pecking

keys for food under tightly controlled conditions, appears to be equally valid for humans conversing in a relatively uncontrolled social setting.

Choice in Novel Combinations

Several investigators have asked whether matching holds for alternatives that have been trained separately. For example, suppose a pigeon has obtained food on a VI schedule by pecking a red key, and has also obtained food on a different VI schedule by pecking a green key, where the experimenter has presented red and green keys successively for fixed periods of time and there is no possibility of choice between them. This procedure is known as a *multiple* schedule, and each stimulus-schedule combination defines a multiple-schedule component. How would the bird's pecks be distributed when red and green are presented together for the first time?

Edmon, Lucki, and Grisham (1980) addressed this question by training pigeons on multiple-schedule components with different VI values on the left and right keys, where the components were presented in strict alternation for 1-minute periods. When baseline response rate in one component was expressed relative to the sum of rates in both components, the response-rate proportion was less extreme than the reinforcer-rate proportion—a common result in multiple schedules (see below). When the two keys were lighted simultaneously in choice probe tests, however, the allocation of responding was more extreme than predicted by matching. Herrnstein and Loveland (1976) also observed overmatching in a novel-pairs probe test after training with various pairs of stimuli in concurrent schedules, and vom Saal (1972) obtained similar results in probe choice tests after training with a discrete-trial procedure. Thus, the matching law may systematically underestimate choice between alternatives that have been experienced separately.

Generalized Matching

Although matching is a common result for steady-state performance on concurrent VI VI schedules, choice allocation sometimes deviates

from the strict matching relation of Equation 1 (see Baum, 1979, for review). Deviations from matching are well described by the expression:

$$B_1/B_2 = c(r_1/r_2)^a, \quad (2)$$

where B_1 and B_2 represent the behavior allocated to alternatives 1 and 2, measured as pecks or time allocation and defined either by key position or the stimulus on the food key; r_1 and r_2 represent the reinforcers for the two alternatives; c represents a bias toward one alternative or the other, such as a position or stimulus preference that is constant across all reinforcement conditions; and a represents the sensitivity of the behavior ratio to the reinforcer ratio. Equation 2 is known as the *generalized matching law* (Baum, 1974), and it describes the relation between ratios of response and reinforcer rates in multiple as well as concurrent schedules (Davison & McCarthy, 1988; McSweeney, Farmer, Dougan, & Whipple, 1986).

The generalized matching law is often expressed in logarithmic form:

$$\log (B_1/B_2) = \log c + a \log (r_1/r_2) \quad (3)$$

because it is easy to estimate the parameters c and a by linear regression of log behavior ratios on log reinforcer ratios. The fitted slope estimates a , and the intercept, where the log reinforcer ratio is 0 (i.e., $r_1 = r_2$), estimates $\log c$. If the ratio of responses exactly equals the ratio of reinforcers, the values of a and $\log c$ will be 1.0 and 0, respectively, and Equation 3 reduces mathematically to strict matching, as found by Herrnstein (1961). When a is 1.0 but $\log c$ deviates from 0, the result is known as *biased matching*. When a is reliably less than or greater than 1.0, the results are termed *undermatching* or *overmatching*, respectively. Some hypothetical data illustrating various values of c and a are shown in Figure 12-2.

The distinction between bias (c) and sensitivity (a) is especially important in applied situations that provide limited data. For example, suppose the experimenter arranges reinforcers in a 3:1 ratio for two alternative responses, and the subject re-

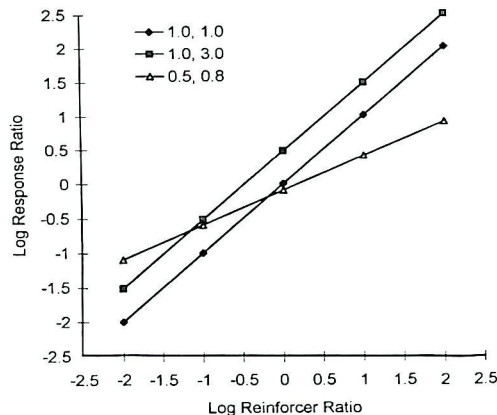


Figure 12-2. Hypothetical choice data illustrating different values of sensitivity and bias in a logarithmic expression of the generalized matching law, Equation 3. Exact matching results when both a and c are 1.0. When a is 1.0 and c is 3.0, the function is shifted upward, representing biased matching. When a is 0.5 and c is 0.8, the function is shallower and shifted downward from the origin (0,0), representing undermatching with a small bias.

sponds in a 2:1 ratio. From this result, one cannot determine whether response allocation is biased away from the more frequently reinforced response, or whether it undermatches the reinforcer ratio. At a minimum, response allocation must be measured at two different reinforcer ratios to estimate c and a , and at least five reinforcer ratios are customarily employed in basic research. When the parameters c and a are estimated reliably, they may be construed as higher-order dependent variables that characterize choice behavior under a given set of experimental conditions, and that may be related in an orderly way to quantifiable aspects of those conditions.

Some Determiners of Bias and Sensitivity in Research on Choice

A great deal of research, summarized by Davison and McCarthy (1988), has attempted to identify the determiners of c and a in Equation 2, and to evaluate their independence. A few examples will be reviewed here. In a two-response proce-

dure, pigeons' choice is strongly biased toward key pecking when the alternative response is lever pressing, although time allocation is less biased, perhaps because lever pressing is more awkward and time-consuming (Davison & Ferguson, 1978). Bias also depends on qualitative differences in the reinforcers provided by the alternatives (e.g., type of grain: Miller, 1976; food vs. brain stimulation: Hollard & Davison, 1971). The biasing effects of response type and reinforcer quality did not affect sensitivity in these examples.

In concurrent switching-key schedules, sensitivity increases when the stimuli defining the alternatives are made more distinctive (Alsop & Davison, 1991; Miller, Saunders, & Bourland, 1980). This is not surprising because, in the limit, sensitivity to reinforcer ratios must be zero when the alternatives are indistinguishable. The distinctiveness of the alternatives may also be enhanced by increasing the effort required to switch between them. For example, in a switching-key procedure, the allocation of time to an alternative became more extreme than matching (i.e., $a > 1.0$ or overmatching) when the number of responses required to switch between alternatives was increased (Pliskoff, Cicerone, & Nelson, 1978). Likewise, in a two-key procedure, strong overmatching ($a \gg 1.0$) resulted when substantial distance and effort were required to travel between the food keys (Baum, 1982). It may be that any variable promoting separation between concurrent alternatives will enhance sensitivity to the reinforcer ratio and lead to overmatching.

Effects of Reinforcement Parameters

The matching law was extended to incorporate the effects of reinforcer amount and immediacy by Baum and Rachlin (1969) and Killeen (1972). A generalized version, in which each variable is expressed as a ratio and raised to a power that characterizes sensitivity to that variable, was suggested by Davison and McCarthy (1988):

$$B_1/B_2 = c(r_1/r_2)^a (m_1/m_2)^b (i_1/i_2)^d \quad (4)$$

where m and i represent the amounts and immediacies of reinforcement, subscripted for the two alternatives. This equation implies that each dimension of reinforcement affects choice independently, and that they combine multiplicatively (or additively in a logarithmic version).

If reinforcer amounts and delays are unequal but remain constant while reinforcer rates are varied, they are mathematically equivalent to the bias term c . Thus, Equation 4 implies that sensitivity to reinforcer ratios, a , will be unaffected by other aspects of reinforcement, and conversely that sensitivity to other variables such as amount will be unaffected by reinforcer ratios.

Several findings have challenged this implication. For example, Davison (1988) found that sensitivity to reinforcer amount decreased as reinforcer rate increased. More generally, Equation 4 implies that choice allocation depends on the ratios of reinforcement variables and not their absolute values. However, Logue and Chavarro (1987) found that response ratios decreased when absolute amounts increased in a constant 3:1 ratio (with reinforcer rates held constant and equal), and that response ratios increased when total reinforcer rate increased with a fixed 3:1 reinforcer rate ratio. Alsop and Elliffe (1988) confirmed the latter result in a parametric study that varied both absolute and relative reinforcer rate: as absolute reinforcer rate increased, sensitivity to reinforcer ratios also increased. Therefore, a full account of choice behavior must consider absolute as well as relative values of reinforcer rate and amount. The achievement of such an account remains a challenge for the future.

Generalized Matching in Concurrent Chains

Systematic deviations from strict matching in concurrent chain VI VI schedules of the sort first studied by Autor (see above) were reported by Fantino (1969). His experiment varied the lengths of the two equal initial links with terminal-link delays of 30 and 90 seconds. Matching was obtained with initial links of moderate length, but when the initial links were very long,

initial-link choice approached indifference between the terminal-link schedules, and when the initial links were short, it approached exclusive preference for the richer terminal link. Thus, matching appears as a point on a continuum, depending on the temporal context in which the terminal links are set.

Grace (1994) has extended the generalized matching law to describe concurrent-chains performance. His model is:

$$B_{i1}/B_{i2} = c(r_{i1}/r_{i2})^{a_i} [(i_{i1}/i_{i2})(x_{i1}/x_{i2})]^{a_i(T_i/T_i)}, \quad (5)$$

where B_{i1} and B_{i2} represent responding in the initial links, r_{i1} and r_{i2} represent the rates of entry into the terminal links arranged by the initial-link schedules (which are usually equal), i_{i1} and i_{i2} represent the immediacies of food reinforcement in the terminal links, x_{i1} and x_{i2} represent other factors that influence terminal-link value such as reinforcer amount, T_i and T_i represent the average durations of the terminal and initial links, and a_i and a_i represent the sensitivities of initial-link allocation to the ratios of initial-link and terminal-link reinforcer rates, respectively. The exponent (T_i/T_i) captures the effects of relative initial-link and terminal-link length on initial-link preference reported by Fantino (1969). Grace (1994) showed that the model provided an excellent account of a wide range of published findings, and Grace (1995a) applied his equation to data on choice between terminal links that differed in both immediacy and amount of reinforcement to show that the effects of these variables were independent.

Is Matching Normative?

In view of all the empirically determined a -values that depart systematically from 1.0, it is hard to maintain the proposition that strict matching is truly general. Nevertheless, strict matching to the relative value of the alternatives may be normative for all situations, with deviations resulting from suboptimal selection of procedural variables by the experimenter or from failures of discrimination by the subject (Baum, 1979; Davison & Jenkins, 1985).

If strict matching is assumed, the parameters of the generalized matching law may be used to scale relative value. For example, consider a hypothetical experiment in which response ratios match reinforcer-rate ratios across a series of concurrent VI VI schedules for a given subject. Then, in the next phase of this hypothetical experiment, reinforcer rates are kept equal and constant while the amounts of food are varied independently for the two alternatives. If the ratio of responses is related to the ratio of reinforcer amounts according to Equation 4, with $b = 0.5$, one could argue that the subjects were matching their choices to the relative value of food, and that relative value is itself a power function of relative amount with an exponent of 0.5. This conjecture could be tested by arranging novel combinations of reinforcer rates and amounts, and using Equation 4 to predict response ratios (for a full discussion, see Killeen, 1972).

Relatedly, matching may be taken as an axiom for theory development. For example, the assumption of strict matching was used by Fantino (1969) to develop the delay-reduction theory of conditioned reinforcement, and thereby to explain the effects of relative initial-link and terminal-link length in concurrent chains (described above). Fantino assumed that subjects matched their initial-link choices to the relative reduction in delay to the reinforcer signaled by onset of the terminal-link stimuli. In equation form,

$$B_{i1}/B_{i2} = (T - t_1)/(T - t_2), \quad (6)$$

where B_{i1} and B_{i2} represent initial-link responses to alternatives 1 and 2, t_1 and t_2 represent the average times from terminal-link onset to food, and T is the overall time between food reinforcers averaged over both links for both alternatives. Thus, $(T - t_1)$ and $(T - t_2)$ represent the reduction in delay to food signaled by the terminal-link stimuli, which Fantino has equated with their conditioned reinforcing value. Delay-reduction theory accounts well for performance in a wide variety of procedures involving signaled delays to reinforcement (see Fantino, Preston, & Dunn, 1993, for a recent review).

The predictions of delay-reduction theory are generally similar to those of Grace's (1994) model (Equation 5). However, delay-reduction theory implies that the conditioned reinforcing values of the terminal-link stimuli depend on the temporal context in which they appear, whereas Grace's model implies that conditioned reinforcing values depend only on the delays to reinforcement in their presence, with temporal context modulating preference between them. Grace's (1995b, Experiment 4) data generally support the independence of value from temporal context. Because applied work often relies on conditioned reinforcers, it is important to understand the determiners of their value.

Theoretical Explanations of Matching

Concurrent VI VI schedules may interact with behavior to produce matching as a byproduct of some more fundamental process. For example, if subjects follow the local probability of reinforcement as it changes through time spent on one or the other schedule (see above), they should switch between VI alternatives in a predictable pattern that closely approximates overall matching. Shimp (1966) examined sequential patterns of responding in a discrete-trial two-key choice experiment that mimicked this aspect of concurrent VI VI schedules, and found that his subjects tended to respond to whichever key had the momentarily higher probability of reinforcement. This pattern of responding, termed *momentary maximizing*, leads to matching on concurrent VI VI schedules (Shimp, 1966; Stadon, Hinson, & Kram, 1981). However, the pattern reported by Shimp has not always been found in related studies (e.g., Heyman, 1979; Nevin, 1969), so the empirical basis for momentary maximizing is less well established than the matching result it was supposed to explain.

Rachlin, Green, Kagel, and Battalio (1976) suggested a different account of matching known as *molar maximizing*. In concurrent VI VI schedules, subjects can obtain more reinforcers by responding to both alternatives than if they re-

spond exclusively to either one; therefore, switching between alternatives tends to maximize total reinforcement. Some plausible assumptions about the relation between the allocation of responding to each alternative and the number of reinforcers obtained from that alternative, known as the *molar feedback function* for VI schedules (see below), lead theoretically to matching as the allocation that maximizes total reinforcement (e.g., Baum, 1981; Staddon & Motheral, 1978). However, experiments designed specifically to pit matching against molar maximizing have usually found that the allocation of behavior roughly matches relative obtained reinforcement even when matching leads to a substantial loss of total reinforcement (e.g., Mazur, 1981; Vaughan, 1981).

Herrnstein and Vaughan (1980) have proposed a third account of choice behavior, termed *meliorizing*, that emphasizes the local rates of reinforcement, defined as reinforcers obtained per unit time spent engaged in each alternative. Response allocation is assumed to shift toward the alternative with the greater local reinforcer rate, and becomes stable when neither alternative has a greater local reinforcer rate than the other, that is:

$$r_1/t_1 = r_2/t_2, \quad (7)$$

where t_1 and t_2 represent the times spent on the two alternatives. (Note that Equation 7 implies strict matching.)

Meliorizing accounts for performance on concurrent VR VR and VI VR as well as VI VI schedules. However, some predictions of meliorizing have been disconfirmed by Williams and Royalty (1989). They presented alternatives that differed in both local and overall reinforcer rates in probe tests with novel pairs, and found that response allocation was based on overall rather than local rates of reinforcement.

In a review of recent theoretical developments in the study of choice, Williams (1994) concluded that "none of the major theories can presently explain all of the available data" (p. 105). However, the failure of any theory to explain all of the data should not lead applied re-

searchers to dismiss the issues raised by theoretical analyses. For example, momentary maximizing theory calls attention to the importance of how choice behavior interacts with moment-to-moment schedule contingencies, and molar maximizing theory does likewise for long-term outcomes. Meliorizing provides a way to understand failures of molar maximizing and thus is directly relevant to real-world problems that involve some incompatibility between short-term satisfaction and long-term optimality. In effect, meliorizing subjects treat the alternatives as mutually exclusive and choose one or the other on the basis of current differences in their value, rather than choosing between different distributions of behavior to both alternatives on the basis of total value. Thus, meliorizing may help us to understand behavior in situations where choice of a small immediate reinforcer entails the loss of more highly valued but delayed outcomes—situations that are said to require "self-control" and have obvious relevance to many social and clinical problems (see Chapter 13).

Single-Alternative Choice

Many experimental analyses of operant behavior define only one response class, such as lever pressing or key pecking, and examine the rate of responding as a function of the schedule of reinforcement. Herrnstein (1970) suggested that even when only one defined response is eligible for reinforcement, the situation may still be construed as involving choice between that response and unmeasured activities (such as scratching or defecating), designated B_O , that are maintained by naturally occurring reinforcers (such as relief of discomfort), designated r_O . Herrnstein assumed that behavior was allocated between the experimentally defined response B , reinforced according to some schedule at a rate r , and B_O , maintained by r_O , so as to match relative obtained reinforcement. In the form of Equation 1,

$$B/(B + B_O) = r/(r + r_O). \quad (8)$$

Herrnstein further assumed that the sum of all available activities, $B + B_O$, was a constant, des-

ignated k and measured in units of the defined response. Inserting the assumed constancy into Equation 5 and rearranging,

$$B = kr/(r + r_0). \quad (9)$$

According to Equation 9, which is known as Herrnstein's hyperbola or the relative law of effect, the rate of the defined response depends on the rate of reinforcers obtained by that response relative to all reinforcers, including unspecified reinforcers occurring in the situation.

As we have seen, many studies of concurrent operants have challenged the key assumption of strict matching, and several findings have challenged the constancy of k (e.g., McDowell & Wood, 1984). Also, Equation 9 breaks down at very high reinforcer rates or in long experimental sessions (Baum, 1993; McSweeney, 1992). However, it describes response rates maintained by VI schedules in studies with pigeons, rats, monkeys, and humans quite closely in most situations. It also accounts well for the effects of the magnitude of both positive and negative reinforcers in discrete-trial as well as free-operant procedures, and thus provides a good first approximation to a general law of behavior (see de Villiers, 1977, for review and discussion).

Expressions that are mathematically similar to Equation 9 can be derived from other sets of assumptions (e.g., Catania, 1973; Killeen, 1994; Staddon, 1977), but whatever its derivation, Equation 9 captures three important aspects of operant behavior. First, response rate is a monotonically increasing, negatively accelerated function of reinforcement rate or amount, exemplifying diminishing returns as response rate nears k , its asymptote. Second, the rate of approach to asymptote depends on the availability of other reinforcers in the situation. And third, response rate maintained by a given schedule may be increased or decreased by changing the rate of reinforcement from other sources. These features of Equation 9 are illustrated in Figure 12-3 and should be appreciated by anyone concerned with performance in clinical settings, schools, or the workplace.

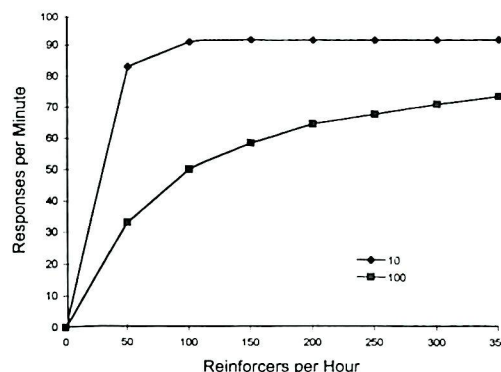


Figure 12-3. Hypothetical data showing the relation between response rate and reinforcer rate according to Herrnstein's hyperbola (Equation 9). When $r_0 = 10$, the function reaches its asymptote at $k = 100$ more rapidly than when $r_0 = 100$. Note that an increase in r_0 from 10 to 100 reduces responding substantially more at low than at high rates of reinforcement.

Response Rate in Concurrent and Multiple Schedules

Equation 9 can be written separately to give the response rates of two concurrent operants, where experimentally specified reinforcers are arranged for each operant:

$$\begin{aligned} B_1 &= kr_1/(r_1 + r_2 + r_0), \text{ and} \\ B_2 &= kr_2/(r_2 + r_1 + r_0). \end{aligned} \quad (10)$$

If these separate expressions for response rate are divided, k and r_0 cancel out, leaving a ratio version of the matching law:

$$B_1/B_2 = r_1/r_2.$$

Thus, matching in concurrent schedules may be seen as the outcome of a process whereby the rate of each response is determined by the reinforcers for that response relative to all concurrently available reinforcers in the situation.

Herrnstein (1970) suggested that if two operants are arranged successively, as in the components of multiple schedules, Equation 10 may be written:

$$\begin{aligned} B_1 &= kr_1/(r_1 + mr_2 + r_0), \text{ and} \\ B_2 &= kr_2/(r_2 + mr_1 + r_0), \end{aligned} \quad (11)$$

where B_1 and B_2 represent response rates and r_1 and r_2 represent reinforcer rates in Components 1 and 2, r_0 represents the rate of unmeasured reinforcers, and m is a parameter taking values from 0 to 1.0 that characterizes the extent to which reinforcers in one component affect responding in the other component. Although Equation 10 fails in some situations (see Williams & Wixted, 1986, for review), it predicts that if $m < 1.0$, response-rate ratios undermatch reinforcer-rate ratios, which is the usual result. It also predicts, correctly, that response rate in one component of a multiple schedule depends on its rate of reinforcement relative to that in the alternated component (for summary, see Nevin, 1973). Thus, as for each alternative in a concurrent schedule, response rate in a multiple-schedule component depends on the context of reinforcement in which it is set.

RESISTANCE TO CHANGE

A Representative Experiment

Multiple schedules are especially useful for the study of resistance to change because components are presented at times and in an order chosen by the experimenter, independently of the subject's behavior, and two or more performances can be compared within subjects and sessions. For example, Nevin (1974, Experiment 1) trained pigeons to peck a single key for food reinforcers, where the key was lighted either red or green for 1-minute periods. When the key was red, a VI schedule arranged 60 reinforcers per hour, and when the key was green, an independent VI schedule arranged 20 reinforcers per hour. Components were separated by a 30-second time-out. After 30 sessions of baseline training, responding was disrupted in brief probe tests of resistance to change by presenting food at various rates, independently of responding, during the time-out periods between components. Response rates during resistance tests were expressed as proportions of baseline. The results are summarized in Figure 12-4, with proportions of baseline plotted on a logarithmic scale. This has the effect of mak-

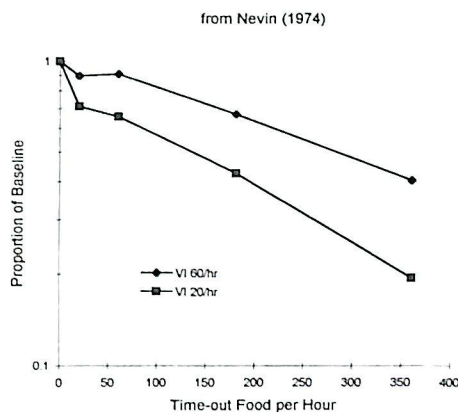


Figure 12-4. Average data for four pigeons trained on multiple VI schedules arranging 60 and 20 reinforcers per hour in two alternating components, showing the effects of presenting food during time-outs between components in probe tests. At every time-out food rate, response rate decreased less, relative to baseline, in the richer component. Note that the y-axis is logarithmic. Adapted from Nevin (1974, Experiment 1).

ing the functions roughly linear; the ratio of the slopes of these functions can be used to quantify relative resistance to change (Nevin, 1992b; Nevin, Mandell, & Atak, 1983). For present purposes, it suffices to note that at every time-out food rate, the decrease was smaller, relative to baseline, in the richer multiple-schedule component. Nevin (1974, Experiment 3) also found that resistance to time-out food was positively related to reinforcer duration in components with the same reinforcer rate. Thus, it appears that rate and amount of reinforcement have similar effects on resistance to change as well as on choice behavior.

Generality

When pigeons are trained on multiple schedules that differ in reinforcer rate or amount, and resistance to change is evaluated by applying some disruptor equally to the two components, response rate nearly always decreases less, relative to baseline, avoid reforming with the greater rate or amount of reinforcement. For example, Bouzas (1978) trained pigeons on multiple VI VI

schedules yielding 60 and 15 reinforcers per hour in the two components and then introduced a VI schedule of punishment that superimposed 120 electric shocks per hour in both components. He found a smaller relative reduction in the richer component. The same result has been obtained with signaled unavoidable shocks (Lyon, 1963), with pre-session feedings (Eckerman, 1968) or variations in body weight (Herrnstein & Loveland, 1974), with increases in the number of responses required for reinforcement (Elsmore, 1971), and with the discontinuation of reinforcement in both components (extinction—Nevin, 1974, Experiment 2; Nevin, 1988; see Chapter 11). Comparable findings have been obtained with rats, monkeys, and humans (see below and Nevin, 1979).

Research with single schedules has yielded some similar results. For example, Church and Raymond (1967) trained one group of rats on a single VI schedule providing 300 reinforcers per hour, and another on a single VI schedule providing 18 reinforcers per hour. When response rates were disrupted by superimposing a VI schedule of 30 brief electric shocks per hour, there was a substantially smaller reduction, relative to baseline, for the group with the richer VI schedule, a result similar to that of Bouzas (1978) for pigeons on multiple schedules. However, there have been some failures to obtain the usual multiple-schedule results with single schedules. Cohen, Riley, and Weigle (1993; see also Cohen, Furman, Crouse, & Kroner, 1990) have reported consistent failures to find differences in resistance to response-independent food, prefeeding, or extinction in rats or pigeons trained on a variety of single schedules differing in reinforcer rate; indeed, they found that resistance to extinction was often greater for the leaner schedules. In related multiple-schedule experiments, however, they generally obtained greater resistance to change in the richer component. These findings suggest an important role for within-session alternation of stimuli signaling different conditions of reinforcement, and are especially important for applications designed to maximize resistance to change.

Two multiple-schedule experiments with human subjects have confirmed the usual find-

ings of greater resistance in the richer component. Mace, Lalli, Shea, Lalli, West, Roberts, and Nevin (1990) instructed adults with mental retardation to sort two colored sets of plastic dinnerware successively, and arranged a rich VI schedule of popcorn or coffee reinforcers for sorting one color and a lean VI schedule for sorting the other. In effect, the colors defined the components of a multiple schedule. There was no difference in baseline sorting rates, but when a television set was turned on, sorting was less disrupted for the color correlated with the richer schedule. Cohen, (1996) reported a similar result for college students typing at a computer terminal with points toward course credit as reinforcers. A rich VI schedule of points was arranged when the screen was one color, and a lean schedule when it was a different color. Again, there was no difference in baseline typing rates, but typing was more resistant to distraction by an alternative task (finding a cartoon character in a picture book) when the screen signaled the richer schedule. The results for representative individual human subjects are presented in the left column of Figure 12–5, together with results for a representative pigeon. Their similarity suggests that the findings of laboratory research on resistance to change with nonhuman subjects in tightly controlled conditions are valid also for humans in relatively natural settings.

Resistance to Change and the Matching Law

Nevin's (1974, Experiment 1) results, described above and shown in Figure 12–4, are consistent with Herrnstein's (1970) extension of the single-alternative matching law to multiple schedules, Equation 11. If r_{TO} , representing the rate of time-out food, is added to the denominator of the expression for each component, its decremental effect on response rate in a given component, relative to baseline, depends inversely on the rate of reinforcement in that component. More generally, Equation 11 can account at least ordinally for all the resistance findings summarized above if any disruptor reduces the rate of the measured response through increases

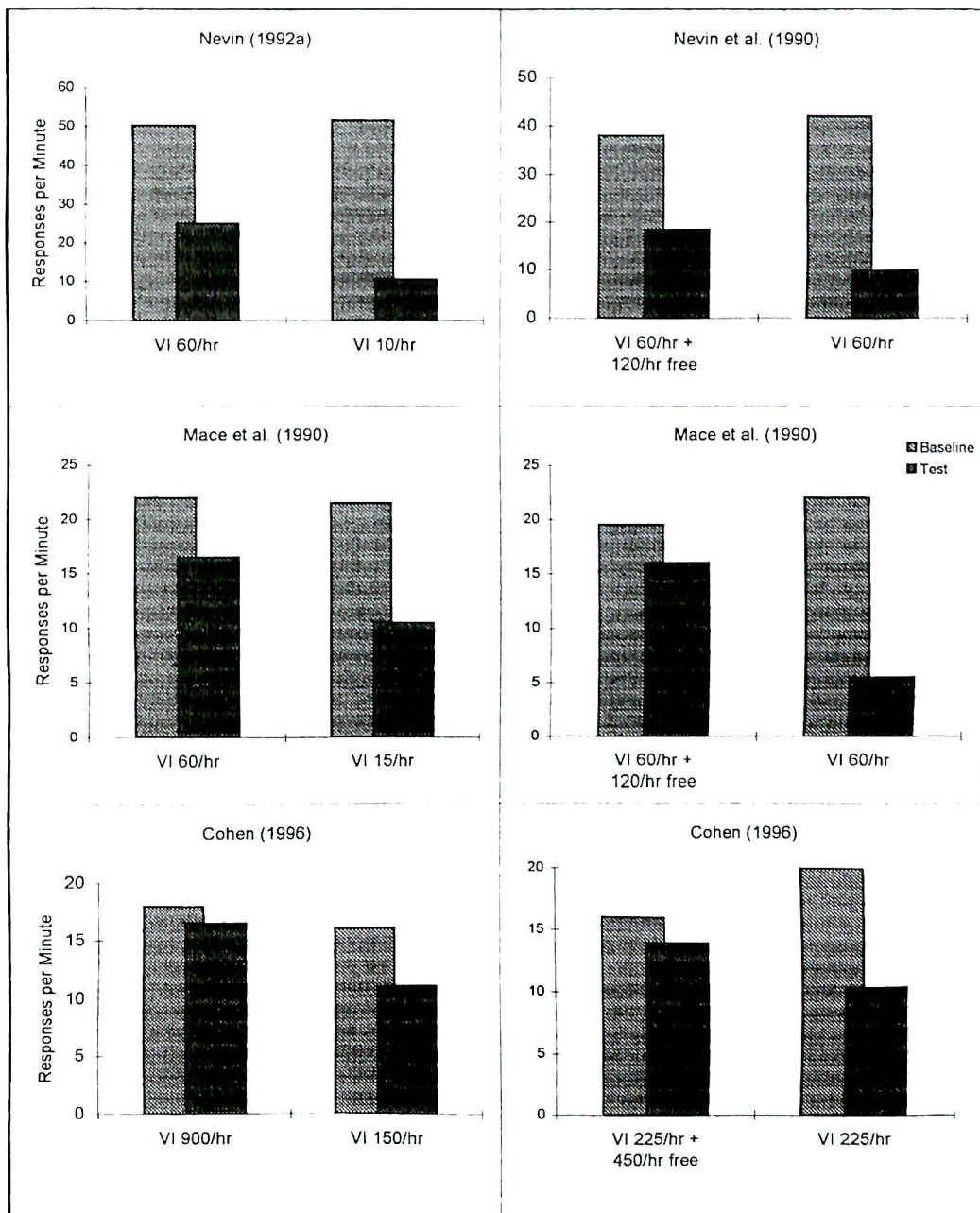


Figure 12-5. Baseline and resistance test data for individual subjects on multiple schedules of reinforcement. The top panels represent key pecking by two pigeons; the center panels represent sorting by S1, an adult with mental retardation; and the bottom panels represent typing by the median undergraduate in two experiments. The left column gives the data for baseline and resistance tests in multiple VI VI schedules where the components differ in VI reinforcer rate. The right column gives the data for baseline and resistance tests in equal multiple VI VI schedules, where free reinforcers have been added to one component. In every case, responding is more resistant to change in the component with the higher total reinforcer rate, regardless of whether all reinforcers were contingent on responding or the ordering of baseline response rates. Adapted from Nevin et al. (1990, Experiment 1); Nevin et al. (1992a, Experiment 2); Mace et al. (1990, Experiments 1 and 2); and Cohen (1996, Experiments 1 and 2).

in some other source of reinforcement, either measured (as in the case of time-out food) or unmeasured (as for punishment or prefeeding). The effects of extinction can be explained similarly by arguing that the relative value of unmeasured reinforcers increases with time since the termination of scheduled reinforcers. Thus, Herrnstein's (1970) formulation, suitably interpreted, can account for many aspects of resistance to change.

However, several findings argue for a separate account. Nevin, Tota, Torquato, and Shull (1990, Experiment 1) arranged identical VI schedules of food reinforcement for pigeons in two multiple-schedule components, and also provided response-independent food on variable-time (VT) schedules in one of those components throughout training. Baseline response rate in the component with added VT food was consistently lower than in the VI-only component, as predicted by Equation 9, since the VT food would contribute to reinforcement for behavior other than key pecking. Another way to explain the lower rate is that added VT food weakens the contingency between pecking and food, in that food is not uniquely contingent on pecking in the VI plus VT component. By contrast, added VT food strengthens the stimulus-reinforcer contingency, in that the stimulus for the VI plus VT component is better correlated with food than the stimulus for the VI-only component. Tests of resistance to prefeeding and to extinction showed that response rate was reduced less, relative to baseline, in the component with added VT food, even though its baseline response rate was lower. This result has been repeated with humans by Mace et al. (1990) and Cohen (1996) in the procedures described above, as shown in the right-hand column of Figure 12-5. These and related findings (Nevin, 1984; Nevin, Smith, & Roberts, 1987) strongly suggest that resistance to change depends on the correlation between stimuli and reinforcers in multiple schedules, independently of the response-reinforcer contingency and the resulting baseline response rate.

The independence of response rate and resistance to change is confirmed by comparing studies by Nevin, Mandell, and Atak (1983) and

Fath, Fields, Malott, and Grossett (1983). Both studies arranged different schedules in two components separated by a time-out. For Nevin et al., the schedules differed in reinforcer rate but maintained similar baseline response rates. For Fath et al., the reinforcer rates were the same, but high or low baseline response rates were established by differential reinforcement of short or long interresponse times in the two components. When resistance to change was tested by presenting time-out food in probe sessions, Nevin et al. found greater resistance in the component with the higher reinforcer rate, as in the results portrayed in Figure 12-4, even though baseline response rates were similar. By contrast, Fath et al. found that resistance to time-out food was similar in the two components, relative to baseline, even though baseline response rates differed markedly (see Nevin, 1992b, for summary of the average data).

Nevin (1992a) asked whether resistance to change depends on the context of reinforcement in the same way as response rate does, and at the same time tested the application of Herrnstein's account of response rate in multiple schedules (Equation 11) to resistance to change. Pigeons were trained on multiple VI VI schedules where the rate of reinforcement in one component was constant while the rate of reinforcement in the alternated component was either richer or leaner in different experimental conditions. Prefeeding and extinction tests showed that response rate in the constant component was more resistant to change when the alternated component was leaner than when it was richer; the average data for extinction are shown in Figure 12-6. This finding provides another instance of the effects of the context of reinforcement. It is also exactly opposite to the predictions of Equation 11 and related formulations, strongly suggesting the need for a separate account of resistance to change (see Nevin, 1992a, for the full argument).

Nevin's (1992a) resistance data are ordinarily related to the stimulus-reinforcer contingency for the constant component, expressed as the ratio of the reinforcer rate in that component to the average reinforcer rate for the entire session (Gibson, 1981). Nevin (1992b) reviewed a number of

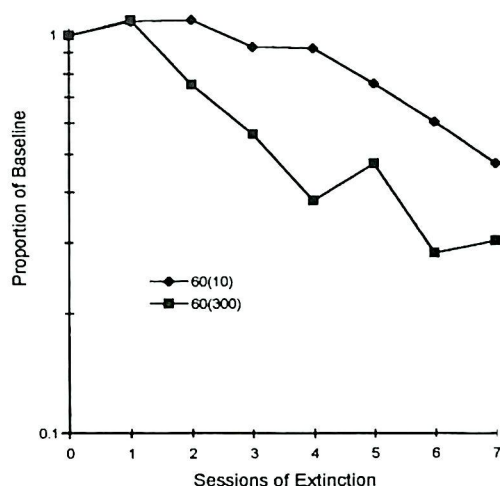


Figure 12-6. Average data for five pigeons trained on multiple VI VI schedules, showing the effects of extinction on responding in a component previously correlated with 60 reinforcers per hour after training in two conditions. Before extinction, there were 10 reinforcers per hour in the alternated component in one condition, and 300 reinforcers per hour in the second. Response rates are expressed as proportions of baseline. For all five subjects, extinction in the constant component progressed more rapidly when the alternated schedule had been rich than when it had been lean. Note that the y-axis is logarithmic. Adapted from Nevin (1992a, Experiment 1).

studies and found that the resistance ratio, measured as the inverse ratio of the slopes of the functions relating log response rate to the external disruptor in two multiple-schedule components, approximated a power function of the ratio of reinforcer rates in those components. The model is similar in mathematical form to the generalized matching law relating the ratio of steady-state response rates to the ratio of reinforcer rates (Equation 2), and to its extension to choice in concurrent chains (Equation 5).

Resistance to Change and Choice

The need for separate accounts of steady-state choice and resistance to change should not obscure the relations between them. In addition to their common mathematical description, initial-link preference in concurrent-chain schedules and

relative resistance to change in multiple schedules depend on similar variables. For example, the terminal-link stimulus with the higher rate of reinforcement is preferred in concurrent chains, and response rate in the multiple-schedule component with the greater rate of reinforcement is more resistant to change, where both results are independent of response-reinforcer contingencies (Neuringer, 1969; Nevin et al., 1990). In a procedure that involved both concurrent-chain and multiple schedules, Grace and Nevin (1997) evaluated preference and resistance to change within subjects and sessions, and found a high correlation between these independent measures. Thus, preference and resistance to change may be convergent measures of the value of a signaled schedule of reinforcement.

THE METAPHOR OF BEHAVIORAL MOMENTUM

In classical physics, the momentum of a moving body is given by the product of its velocity and mass. When an external force is imposed, the change in velocity depends on mass according to Newton's second law:

$$\Delta v = f/m, \quad (12)$$

which implies that for a given external force, the change in velocity is inversely related to mass.

In behavior analysis, reinforcement may endow responding with something analogous to momentum, where baseline response rate under constant conditions is analogous to velocity, and the resistance to change of that response rate is analogous to mass: the smaller the change in response rate when a disruptor is presented, the greater the behavioral mass.

In physics, the velocity and mass of a moving body are independent dimensions, and as we have seen, response rate and resistance to change are independently related to response-reinforcer and stimulus-reinforcer contingencies, respectively. Physical mass is independent of the magnitude of the imposed force, and Nevin et al. (1983) showed that relative behavioral mass in two multiple-schedule components was invariant with respect to the magnitude of the disruptor,

and was internally consistent across three schedule pairs. The momentum metaphor may therefore be useful for quantifying the effects of signaled schedules of reinforcement, some of which may not be evident until behavior is challenged in some way.

Behavioral momentum—the product of baseline response rate and its resistance to change—captures some of the commonsense meaning of “learning.” For example, a child is said to have learned arithmetic well if the child responds rapidly and accurately on arithmetic problems in class (high velocity), and also performs well at home while distracted by TV, or on an achievement test several years later (high mass). However, if the child performed poorly during distraction or on a subsequent test, one would not say that arithmetic was well learned no matter how rapidly and accurately the child performed in class.

Clinicians are often faced with the need to teach desirable behavior so that it occurs reliably during therapy and persists effectively in the client’s natural world after therapy ends—that is, to establish a high level of momentum. Conversely, clients often engage in undesirable behavior that is remarkably persistent, suggesting a high level of momentum based on some combination of circumstances that may be identified through functional analysis (Neef & Iwata, 1994). The momentum metaphor, and an understanding of the separate determiners of behavioral velocity and mass, may therefore be valuable in clinical settings.

APPLICATIONS OF CHOICE AND MOMENTUM ANALYSES

Choice

One goal of applied work is to promote choice of socially desirable behavior over undesirable behavior. Laboratory research on choice is directly relevant to situations in the natural world where people engage in repeated choices between alternatives that are analogous to concurrent schedules. However, context effects of the sort described above are likely to complicate pre-

dictions in applied work, because both desirable and undesirable behavior may be reinforced outside the therapy setting.

Another difficulty for applied work is that real-world contingencies rarely mimic concurrent schedules in detail. However, they can at least be classified as ratio-like or interval-like by examining the relation (known as the molar feedback function) between the rate of responding and the resulting rate of reinforcement. For example, the number of berries obtained by searching in a rich patch increases in proportion to the rate of searching, even though searching does not always reveal berries. This is a ratio-like contingency. By contrast, fishing in a sparsely populated inlet is interval-like: one must cast at least occasionally to catch anything at all, but the total catch is limited by the stock of fish, regardless of how often the line is cast. The molar feedback functions for these types of situations are shown in Figure 12-7.

If both alternatives are ratio-like and lead to the same reinforcer (e.g., money), research and theory on choice behavior suggest that subjects will respond exclusively to the richer alternative. If they lead to qualitatively different reinforcers, though, subjects may switch between alternatives as their reinforcer preferences change. For example, smoking pays off on a very rich schedule—every puff brings smoke into the mouth and lungs. However, the decrease in enjoyment from smoking successive cigarettes, one right after another, ensures that even a heavy smoker will refrain from time to time and engage in some other behavior, such as going to a movie, even though it does not pay off so frequently or reliably.

If both alternatives are interval-like, some approximation to matching can be expected, with biases depending on a host of variables that may be difficult to evaluate in natural settings. Overmatching is likely if the alternatives have been experienced in separate contexts. Thus, the matching law may not always help to predict choice in natural situations.

In controlled situations, however, context effects can be avoided, and concurrent VI VI schedules with different responses and rein-

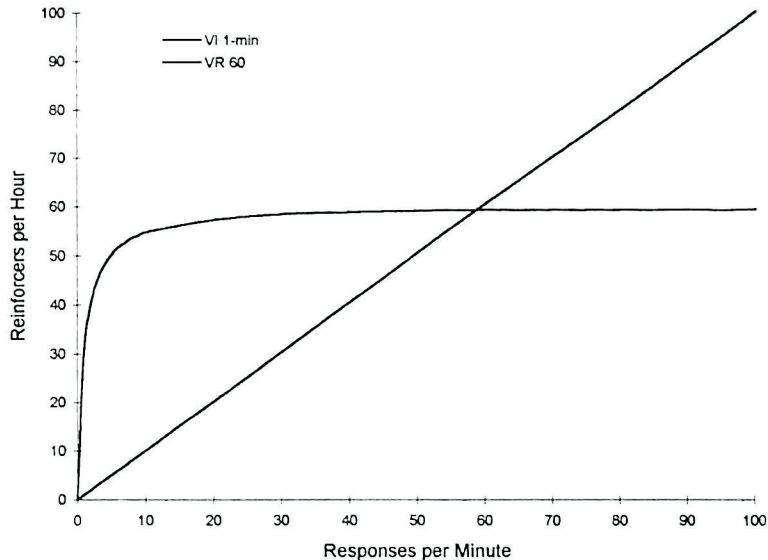


Figure 12-7. Molar feedback functions showing obtained reinforcer rate as a function of response rate. For a VI 1-minute schedule, obtained reinforcer rate rises rapidly toward its asymptote of 60 per hour and is essentially constant for all response rates above 20 per minute. By contrast, for a VR 60 schedule, reinforcer rate increases linearly with response rate.

forcers can be programmed explicitly. For example, Neef, Shade, and Miller (1994) arranged concurrent VI VI schedules with emotionally disturbed children for two sets of arithmetic problems to evaluate the effects of response effort, reinforcer delay, and reinforcer quality. They found large individual differences in the relative effectiveness of these dimensions, and the frequent occurrence of exclusive responding to one or the other problem set precluded quantitative assessments of bias, sensitivity, and independence of dimensions via the generalized matching law, Equation 2. In a similar study with teenagers in a special education program, Mace, Neef, Shade, and Mauro (1994) varied only the concurrent VI VI schedule values and obtained reasonably orderly undermatching. However, a variety of additional procedures was needed to prevent perseveration of response patterns when the schedule values changed. Thus, the practical value of concurrent schedules for quantifying factors that affect choice in individual clients may be limited.

On the other hand, clinical populations may exhibit consistent deviations from normal performance on concurrent schedules, and the generalized matching law can be used to identify and quantify them. For example, Oscar-Berman, Heyman, Bonner, and Ryder (1980) studied the performances of Korsakoff's patients and normal subjects on concurrent VI VI schedules, and observed that Korsakoff's patients were far less sensitive ($a \ll 1.0$ in Equation 2) to variations in reinforcer ratios. Reduced sensitivity to reinforcer ratios may indicate that the patients failed to discriminate the alternatives, or that they scaled reinforcer rate in a different way from normal individuals; either interpretation could contribute to the analysis of Korsakoff's syndrome.

Alternative Reinforcement

One of the difficulties in eliminating unwanted behavior is that the reinforcer may be unknown, or extremely difficult to discontinue. For example, a child's tantrums in a public place

are virtually certain to attract parental attention. Therefore, a common procedure for reducing unwanted behavior is to arrange concurrent reinforcement for a desirable response, or to provide reinforcers if the undesired response does not occur for some time. Equation 9 implies that response rate goes down when alternative reinforcers are added to the denominator, whether or not they are contingent on any specified response. McDowell (1982) used this approach to decrease the oppositional behavior of an emotionally disturbed boy, and showed that the results were consistent with Equation 9.

On the basis of Equation 1 for strict matching and the feedback functions for the undesired and alternative responses, Myerson and Hale (1984) argued that to be most effective, the alternative reinforcer should be substitutable for that maintaining the undesired behavior and should be introduced on interval-like contingencies. However, the practical utility of these prescriptions for alternative reinforcement derived from basic research on steady-state choice behavior may be limited by considerations of behavioral momentum.

Resistance to Change

When alternative behavior is reinforced concurrently in order to reduce the frequency of an undesired response, there is likely to be a concomitant increase in the total rate of reinforcement in the situation. Because resistance to change increases with the total rate of reinforcement in a situation, reinforcing alternative behavior may inadvertently increase the persistence of undesired behavior. Mace (1991) explored two ways to reduce food stealing by a boy with mental retardation. When attempts to steal food were blocked (in effect, extinction of food stealing), the boy stopped trying to steal food within four sessions. However, a high rate of stealing recurred after blocking was discontinued. Mace then reinforced appropriate eating to decrease the rate of food stealing to a low level. When the effectiveness of this procedure was tested by discontinuing it and blocking further attempts to

steal food, the boy persisted in theft attempts for over 30 sessions, in accordance with the predictions of behavioral momentum. The data are shown in Figure 12-8.

The increase in resistance to change produced by added reinforcement is, of course, advantageous if the behavior of interest is desirable. For example, Tota-Faucette (1991) used auditory biofeedback to teach children to relax in two distinctive stimulus situations that were presented alternately as components of a multiple schedule. In the first, the children received points that could be exchanged for toys contingent on meeting a relaxation criterion; in the second, there was an identical schedule of contingent points, and free points were awarded periodically. During training, relaxation levels were similar in the two situations, but relaxation was more resistant to extinction in the second (added free points) situation when auditory feedback and points were discontinued. Both Mace's and Tota-Faucette's studies show that persistence can be enhanced markedly by adding reinforcers that are not contingent on the response of interest.

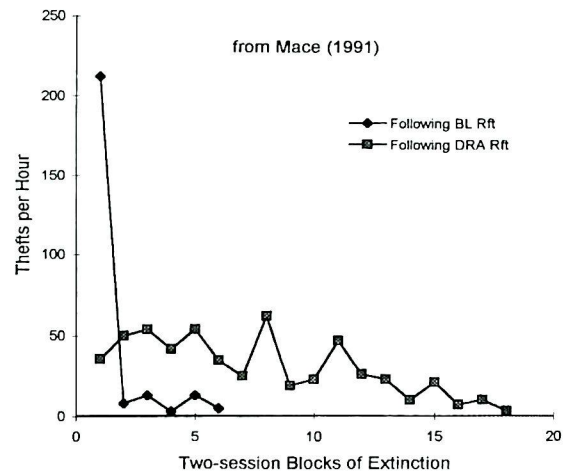


Figure 12-8. The rate of food-theft attempts by a boy with mental retardation when thefts were blocked after baseline (BL) and after differential reinforcement for appropriate mealtime behavior (DRA). Although DRA sharply reduced the theft rate, it also increased its persistence. From Mace (1991).

Behavioral Momentum and Compliance

All the research on behavioral momentum reviewed established a stable baseline in training before assessing resistance to change. However, the stable baseline may not be necessary. Skinner (1938) obtained substantial resistance to extinction after reinforcing a single response, and it may be that one or a few reinforcers are sufficient, metaphorically, to get behavior moving and endow it with enough mass to make its motion persist. Mace, Hock, Lalli, West, Belfiore, Pinter, and Brown (1988) applied this notion to the problem of noncompliance in adults with mental retardation. They presented a few requests that their clients were very likely to obey ("high-*p* commands," such as "Give me five"), and then issued a request that was normally resisted (a "low-*p* command" such as "Empty the trash"). They found much higher rates of compliance with low-*p* commands that followed immediately after a series of high-*p* commands than with low-*p* commands presented alone, suggesting that prompting and social reinforcement for easy instances of compliance generated some momentum for compliance in general. However, the effect was short-lived: there was no evidence of enhanced compliance if low-*p* commands were presented 20 seconds after the high-*p* command sequence.

The persistence of compliance with low-*p* commands can be enhanced by using food rather than praise to reinforce compliance with the preceding high-*p* commands (Mace, Mauro, Boyajian, & Eckert, 1997). Another variable that should be explored is the length of the high-*p* sequence: by analogy with the effects of length of training on resistance to free food in pigeons (Lentz & Cohen, 1980), low-*p* compliance should become even more persistent if high-*p* compliance was reinforced repeatedly. Finally, compliance with low-*p* commands might be especially persistent in an environment that included other reinforcers. For example, Davis, Brady, Hamilton, McEvoy, and Williams (1994) used the high-*p* command procedure to establish

social interaction in children with autism, and found that the level of interaction was maintained after the procedure was discontinued, probably because of the natural reinforcers derived from interaction after it occurred reliably. Thus, extension of the behavioral momentum metaphor to compliance with requests has led to an effective method for establishing and maintaining desirable behavior (see Nevin, 1996, for a more comprehensive discussion).

FUTURE DIRECTIONS

Drug addiction is one of the most serious problems facing the United States and many other nations. Addiction leads to staggering social costs that result from lost productivity, poor health, crime, and personal tragedy. Therefore, the analysis and treatment of addictive behavior have a high priority.

Addiction involves repeated choices between two broad classes of behavior: seeking and using a drug, or engaging in activities that lead to nondrug reinforcers. Herrnstein and Prelec (1992) and Heyman (1996) have suggested that addiction can be understood as an instance of melioration—choosing the momentarily more valued alternative—even though that choice leads to a shifting mix of drug and nondrug reinforcers that is less than optimal, and ultimately catastrophic. Their analysis assumes that the value of any reinforcer is bitonic with the frequency of use: up to a point, value increases with frequency of engagement, as in learning to play a musical instrument, or skiing, or savoring a good wine. After some point, though, value diminishes with additional time spent practicing or skiing or wine-tasting. The same is true with drugs of abuse: with repeated consumption, tolerance develops and the absolute value of a given dose decreases. However, drugs differ from conventional reinforcers not only in their remarkable potency but also in that repeated drug use decreases the values of nondrug reinforcers derived through work, recreation, and social life. Thus, even as their absolute value decreases, the relative value of drugs increases

with repeated use, and the addict eventually becomes trapped into the maximum possible frequency of drug-related behavior. The situation is functionally equivalent to that confronting a pigeon in a two-key procedure where each choice of the richer food schedule leads to a reduction in the overall availability of food (e.g., Vaughan, 1981): persistent meliorizing leads to the worst possible outcome in terms of the total rate of reinforcement.

Although the foregoing analysis is hypothetical, it accords well with the personal accounts of drug addicts (see Heyman, 1996, for review) and thus suggests that research on behavioral methods for bringing overall choice allocation under the control of its long-term consequences, rather than the momentary outcome of each successive choice, could be invaluable for the health and well-being of countless individuals. In applied work with nonaddicted drug users, one approach may involve training to limit the consumption of drugs—or any reinforcers, for that matter—and then to establish environment-reinforcer relations that make a pattern of moderation highly resistant to change.

For recovering addicts who are abstinent, and for those who have never used drugs, the goal is to make abstinence highly resistant to the temptations that abound in a drug-using society. Although a good deal is known about how to establish well-specified behavior so that it persists in the face of a disruptive challenge, much less is known about the persistence of the *nonoccurrence* of some class of behavior—metaphorically, establishing high inertial mass for refraining from some action. There is an urgent need for the extension of basic research on choice and momentum to help develop effective strategies for treating and preventing addiction.

ENDNOTE

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